THE COHERENCE DECOMPOSITION

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Abstract. We introduce two new invariants of the fractional field $\mathcal{F}(n) = \{k/n : 1 \le k \le n-1\}$: the *pairwise alignment* $\sigma(n)$ and the *focused alignment* $F(n) = \alpha(n) - \sigma(n)$, where $\alpha(n)$ is the repetend alignment of previous papers. For digit-partitioning primes, these admit exact formulas:

$$\alpha(ps) = \frac{2s-1}{ps-1}, \qquad F(ps) = \frac{s}{ps-1}, \qquad \sigma(ps) = \frac{s-1}{ps-1},$$

with limits $2/p$, $1/p$, and $1/p$ respectively as the smooth factor $s \to \infty$.

The decomposition $\alpha = F + \sigma$ separates total coherence into a focused component (how strongly $1/n$ is distinguished within its field) and a diffuse component (the background pairwise coherence among all fractions). For digit-partitioning primes, the golden threshold $1/\varphi$ lies strictly between the focused and total limits if and only if $p \in \{2, 3\}$. Among primes $p \ge 3$, only $p = 3$ has this property.

1. Introduction

This paper decomposes alignment into focused and diffuse components.

The preceding papers in this series studied the repetend alignment $\alpha(n)$, which measures the coherence of the fractional field $\mathcal{F}(n)$ relative to $1/n$. This paper introduces a complementary measure: the *pairwise alignment* $\sigma(n)$, which quantifies the background coherence among all pairs of fractions in the field, independent of any distinguished member.

The difference $F(n) = \alpha(n) - \sigma(n)$ isolates the *focused alignment*: the excess coherence that $1/n$ enjoys over a generic field member. Together, α, σ, and F decompose the coherence of $\mathcal{F}(n)$ into structural components with distinct limits and distinct roles in the golden ratio classification.

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2. Definitions

Definition 1. For $n \geq 3$, the *pairwise alignment* of n in base b is

$$\sigma(n) = \binom{n-1}{2}^{-1} \sum_{1 \leq j < k \leq n-1} a(j/n,\, k/n).$$

where $a(j/n,\, k/n)$ is the position-wise digit-match proportion between the repetends of j/n and k/n. Pairs of terminating fractions contribute $a = 1$. Pairs of non-terminating fractions with the same cycle length are compared position-wise. All other pairs (terminating with non-terminating, or different cycle lengths) contribute $a = 0$.

Definition 2. The *focused alignment* of n is

$$F(n) = \alpha(n) - \sigma(n).$$

Focused alignment measures how much more coherent the field is with $1/n$ than with a randomly chosen pair of its own members.

3. The Decomposition for Digit-Partitioning Primes

Theorem 3. *Let p be a digit-partitioning prime in base b and let $s \geq 1$ be a b-smooth integer. Then*

$$\alpha(ps) = \frac{2s-1}{ps-1}, \tag{1}$$

$$F(ps) = \frac{s}{ps-1}, \tag{2}$$

$$\sigma(ps) = \frac{s-1}{ps-1}. \tag{3}$$

Proof. Equation (1) is the alignment formula from [1].

For σ, we count the pairs (j, k) with $j < k$ and $a(j/n, k/n) = 1$. There are three sources of such pairs.

Terminating–terminating pairs. The $s - 1$ terminating fractions (multiples of p in $\{1, \ldots, ps-1\}$) all contribute $a = 1$ in pairs by the definition of pairwise alignment, giving $\binom{s-1}{2}$ pairs.

Identical-repetend pairs among non-terminating fractions. By the cycle-partition argument in [1] (Universal Alignment Formula section), the $(p-1)s$ non-terminating fractions in $\{k/(ps)\}$ partition into $(p-1)/L$ families of sL each, with $L = \mathrm{ord}_p(b)$. Within each family, the fractions split into L same-repetend classes of s each, indexed by the cyclic shift of the base repetend. (Each of the L shifts is achieved by exactly s fractions, since the residue $r \in \{1, \ldots, p-1\}$ producing a given shift has

s representatives in $\{k : 1 \le k \le ps-1,\ p \nmid k\}$.) Across all $(p-1)/L$ families, there are $(p-1)/L \cdot L = p-1$ same-repetend classes total, each of size s. Within each same-repetend class, all $\binom{s}{2}$ pairs have identical repetends and contribute $a = 1$. The total is $(p-1)\binom{s}{2}$ pairs.

All other pairs contribute 0. Pairs of non-terminating fractions with different repetends (i.e., from different same-repetend classes) are cyclic shifts of each other; by the digit-partitioning characterization in [1], the digit function is injective on $\{1, \ldots, p-1\}$, so distinct cyclic shifts share no positions and contribute $a = 0$. Pairs mixing a terminating with a non-terminating fraction contribute $a = 0$ by definition.

Summing:

$$\Sigma = \binom{s-1}{2} + (p-1)\binom{s}{2}.$$

Expanding and dividing by $\binom{ps-1}{2}$:

$$\begin{aligned}\sigma(ps) &= \frac{\binom{s-1}{2} + (p-1)\binom{s}{2}}{\binom{ps-1}{2}} = \frac{(s-1)(s-2)/2 + (p-1)s(s-1)/2}{(ps-1)(ps-2)/2} \\ &= \frac{(s-1)\big[(s-2) + (p-1)s\big]}{(ps-1)(ps-2)} = \frac{(s-1)(ps-2)}{(ps-1)(ps-2)} = \frac{s-1}{ps-1}.\end{aligned}$$

Finally, $F = \alpha - \sigma = (2s-1)/(ps-1) - (s-1)/(ps-1) = s/(ps-1)$. □

Corollary 4. *As $s \to \infty$ through b-smooth integers,*

$$\alpha(ps) \to \frac{2}{p}, \qquad F(ps) \to \frac{1}{p}, \qquad \sigma(ps) \to \frac{1}{p}.$$

At infinite resolution, total coherence splits equally between focused and diffuse components.

4. The Golden Gap

Theorem 5. *For digit-partitioning primes p not dividing b, the golden threshold $1/\varphi$ lies strictly between the focused and total coherence limits, $1/p < 1/\varphi < 2/p$, if and only if $p \in \{2, 3\}$. For $p = 2$, the digit-partitioning formula gives $\alpha(2s) = 1$ identically, so the threshold question is trivial. Among primes $p \ge 3$, only $p = 3$ has the gap.*

Proof. The condition $1/p < 1/\varphi$ requires $p > \varphi \approx 1.618$, satisfied by all primes. The condition $1/\varphi < 2/p$ requires $p < 2\varphi \approx 3.236$, satisfied only by $p = 2$ and $p = 3$. □

For $p = 3$, the limits are $F \to 1/3 \approx 0.333$ and $\alpha \to 2/3 \approx 0.667$, with $1/\varphi \approx 0.618$ falling in between. The golden threshold occupies the gap between focused and total coherence. For $p \geq 5$, the total limit $2/p \leq 2/5 = 0.4$ falls below $1/\varphi$, so both F and α remain below the golden threshold at all resolutions.

This gives a new characterization of the golden ratio's role. The three-tier classification arises because $p = 3$ is the unique prime whose diffuse coherence σ is large enough to push the total $\alpha = F + \sigma$ above $1/\varphi$. The focused component alone never reaches $1/\varphi$ for any prime. It is the accumulation of pairwise background coherence that carries the total across the threshold.

5. The Sigma-Zero Characterization

Proposition 6. *For a prime $p \geq 3$ not dividing b: $\sigma(p) = 0$ if and only if p is digit-partitioning in base b.*

Proof. If p is digit-partitioning, no two distinct fractions j/p and k/p share a digit at any common position, so every off-diagonal $a(j/p, k/p) = 0$ and $\sigma(p) = 0$.

Conversely, if p is not digit-partitioning ($p > b+1$), there exist distinct r, s with $\delta(r) = \delta(s)$. The fractions r/p and s/p share the digit $\delta(r)$ at position 0, giving $a(r/p, s/p) \geq 1/L > 0$ where $L = \mathrm{ord}_p(b)$. Hence $\sigma(p) > 0$. □

This provides an equivalent characterization of the digit-partitioning property: p is digit-partitioning if and only if no two distinct fractions in $\{k/p : 1 \leq k \leq p-1\}$ share a digit at any common position.

6. Composite Denominators

For composite rough parts, the decomposition $\alpha = F + \sigma$ continues to hold by definition, but the exact formulas for digit-partitioning primes do not extend directly. The composite case requires a theory of cross-cycle digit coincidences that is beyond the scope of this paper.

7. Remarks

Three invariants. The fractional field $\mathcal{F}(n)$ now carries three defined invariants:

- $\alpha(n)$: total coherence (alignment with $1/n$),
- $\sigma(n)$: diffuse coherence (pairwise background alignment),
- $F(n)$: focused coherence (structural identity of $1/n$).

These are distinct measures with distinct limits and distinct roles in the classification. Alpha connects to the golden ratio. Sigma characterizes digit-partitioning at prime level. F measures the excess coherence of $1/n$ over a generic pair. All three are computable from the fractional field alone.

The gap as a structural principle. The golden threshold does not act on a single quantity. It acts on the *gap* between focused and total coherence. The three-tier classification arises because $p = 3$ is the unique prime for which this gap is wide enough to contain $1/\varphi$. The width of the gap is $1/p$ (the diffuse limit), and the condition $1/p > 1/\varphi - 1/p$ simplifies to $2/p > 1/\varphi$, recovering the golden bound $p < 2\varphi$.

References

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